



United International University

School of Science and Engineering

Mid-term Examination; Year 2025; Trimester: Summer

Course: BIO 3105/3507; Title: Biology for Engineers; Sec: All

Full Marks: 30; Time: 1 hr 30 mins.

There are Five Questions. 1, 2, and 3 are mandatory to answer, and answer one question from 4 or 5 (anyone).

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| 1. | a. Differentiate Prophase-I of Meiosis and Mitosis Prophase in cell division. | 2 | CO1 |
| | b. How a “normal cell” transformed into an “oncogenic” cell? | 2 | CO1 |
| | c. Mention the characteristic feature of “genetic code”. | 1 | CO1 |
| | d. Define and differentiate “Coding” and “Non-coding” RNAs. | 2 | CO1 |
| 2. | a. As a CSE/DS graduate, design an innovative idea and explain how it could be applied in the biomedical research and development. | 3 | CO2 |
| | b. As a genetic material which one among “DNA” and “RNA” is better? Justify. | 2 | CO2 |
| | c. Illustrate how ribosomes synthesis protein for a variety of purposes. | 3 | CO2 |
| 3. | a. Define “Human Genome Project (HGP)”. Mention the timeline and Goal of the Human Genome Project. | 3 | CO3 |
| | b. How you can determine “Chromosomal deletion”? Describe a disease if large part of a chromosome is deleted. | 3 | CO3 |
| | c. Briefly describe “Proto-oncogenes” and “Tumor suppressor genes” with an example of each. | 2 | CO3 |
| 4. | a. Explain the structure of a “nucleotide”. | 3 | CO4 |
| | b. Comment on “Chromosomal Aberrations”? How Chromosomal aberration can cause “Down Syndrome”. Describe briefly and graphically show that maternal age could be is a factor to develop this disease. | 4 | CO4 |
| 5. | a. Why do we differ from our parents? Remarks on the origins of genetic diversity. | 3 | CO4 |
| | b. Outline “genomic medicine”? Comments on how genomic medicine and personalized therapy could be revolutionary in the coming days. | 4 | CO4 |

CO1: Describe different biological quantities.

CO2: Apply the knowledge of biological systems in a real-life problem.

CO3: Design several biological systems with constraints.

CO4: Explain several procedures for solving biological systems within constraints.